



Comprehensive Notes on Your React Project

This document provides an **in-depth explanation** of every concept used in your React project, including commented-out sections. It covers **file structure**, **JSX**, **functional components**, **conditional rendering**, **event handling**, and **more**, along with **examples and outputs**.

Project File Structure

Your project consists of **four files**:

1. **App.js** - The main application component.
 2. **Test.js** - Handles conditional rendering and list rendering.
 3. **Demo.js** - A simple component.
 4. **Demo2.js** - Another simple component.
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1. App.js - Main Application Component

Concepts Covered

-  Functional Components
-  JSX (JavaScript XML)
-  Expressions inside JSX
-  Event Handling
-  Conditional Rendering with Ternary Operator

Code with Explanation

```
// import logo from './logo.svg';
// import './App.css';
import Test from "./Test"; // Importing Test component

function App() {
  // Example of a JavaScript variable
  let x = 5;
  let text = "Hello Buddy!";

  // If statement use case
  if (x < 10) {
    text = "You right!";
  }

  // Function to handle button click event
  const handleSubmit = () => {
    alert("Hello");
  }
}
```

```

    };

    // Function with parameter
    const handleSubmitWithParam = (message) => {
      alert(message);
    };

    return (
      <div>
        Hi this is Jason
        <Test /> {/* Rendering the Test component */}

        {/* Expressions inside JSX */}
        <h1>React is {5 + 5} times better with JS</h1>
        <input type="text" />
        <p className="note">This is an example of className attribute</p>

        {/* Displaying JavaScript variable inside JSX */}
        {text}

        {/* Ternary Operator for Conditional Rendering */}
        {x < 10 ? "Hello" : "Goodbye"}

        {/* Event Handling Example */}
        <button onClick={handleSubmit}>Submit</button>
        <button onClick={() => handleSubmitWithParam("Click")}>Submit with param</button>
      </div>
    );
  }
}

export default App;

```

2. Test.js - Handling Conditional Rendering & Lists

Concepts Covered

- 👉 Conditional Rendering
- 👉 Using `map()` for list rendering
- 👉 Dynamic Component Rendering

📌 Code with Explanation

```

import Demo from "../Demo"; // Importing Demo component
import Demo2 from "../Demo2"; // Importing Demo2 component

function Test() {
  // Boolean variable for conditional rendering

```

```
let demo = false;

return (
  <div>
    {/* Conditional Rendering - If demo is true, show <Demo />, else show <Demo2 /> */}
    {demo ? <Demo /> : <Demo2 />}

    {/* Mapping Over an Array to Display List Items */}
    <ul>
      {[ "Item 1", "Item 2", "Item 3"].map((item, index) => (
        <li key={index}>{item}</li>
      ))}
    </ul>
  </div>
);
}

export default Test;
```

3. Demo.js - A Simple Functional Component

Concepts Covered

- 👉 Functional Components
- 👉 JSX Syntax

📌 Code with Explanation

```
import React from "react"; // Import React library

function Demo() {
  return (
    <div>
      <p>Demo 1</p>
    </div>
  );
}

export default Demo;
```

4. Demo2.js - Another Functional Component

Concepts Covered

- 👉 Functional Components
- 👉 JSX Syntax

📌 **Code with Explanation**

```
import React from "react"; // Import React library

function Demo2() {
  return (
    <div>
      <p>Demo 2</p>
    </div>
  );
}

export default Demo2;
```

 Summary of Key React Concepts

Concept	Explanation	Example
Functional Components	Simple functions returning JSX	function MyComponent() { return Hello ; }
JSX	Writing HTML inside JS	{5+5}
Conditional Rendering	Displaying different UI based on conditions	{isLoggedIn ? <Welcome /> : <Login />}
Event Handling	Handling user interactions	<button onClick={() => alert("Hi!")}>Click</button>
Lists & .map()	Rendering multiple elements dynamically	{items.map((item) => {item})}

 Final Notes

- **App.js** is the **main entry point** and handles the main UI.
 - **Test.js** contains **conditional rendering and list rendering**.
 - **Demo.js** & **Demo2.js** are **simple components** used for conditional rendering.
 - **Always match import paths to actual filenames** to avoid errors.
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This **comprehensive guide** covers everything in your project with **detailed explanations, examples, and outputs.** 🚀
Let me know if you need further clarifications! 😊